A Collection of Stephan ten Brink's Work

Pin-Jing Li

National Yang Ming Chiao Tung University

ouo.ee11@nycu.edu.tw

Advisor: Prof. Yu-Chih (Jerry) Huang

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A Polar Subcode Approach to BP List Decoding

- Idea: Using Aut of standard polar for the transformed code.
  - Writing CRC-aided polar codes as standard polar subcode.
  - Using LTA as automorphic action to enable parallelism for BP decoding.
- Idea: Relaxing inner polar code. (Not exactly pruning.)
- Experiment: Comparison on different permutation used.
  - Draw one permutation per frame
  - LTA being slightly better than default graph.
  - stage shuffle and GA group are degrading the performance.
  - Explanation: LTA is aut of the inner polar code.
- Result: BP list decoding on CRC aided polar code.
  - on $(128, 64)$ and $(256, 128)$ polar-CRC8, with list size 8.
  - Latency-SNR performance compared with SCL8.

Post Transformation

CRC-aided polar code can be represented as a polar subcode

$$\mathbf{H} = \mathbf{V}\mathbf{G}_N^T, \quad \mathbf{G} = \mathbf{W}\mathbf{G}_N$$

and is further equivalent considering invertible transformation $\mathbf{T}$

$$\mathbf{G} = \mathbf{G}\mathbf{T}\mathbf{T}^{-1} = (\mathbf{G}\mathbf{T}\mathbf{G}_N^{-1})\mathbf{G}_N\mathbf{T}^{-1} = \mathbf{W}_\mathbf{T}\mathbf{G}_N\mathbf{T}^{-1}$$

(Yes, it is exactly the same.)

Decoding process:

1. draw $\mathbf{T}$s from $\mathrm{Aut}(\mathbf{G}_N)$ (they choose LTA here).
2. permute the received signal with $\mathbf{T}$
3. decode as a codeword from $\mathbf{W}_\mathbf{T}\mathbf{G}_N$
4. CRC check

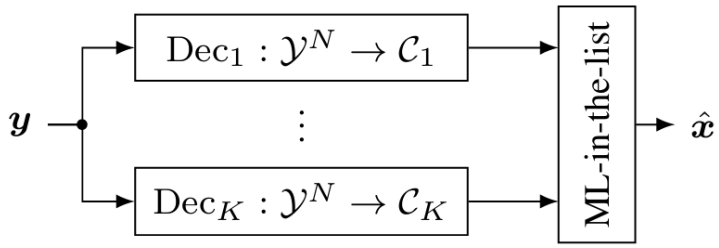
Relaxation

Relaxation is collected as a linear operation as

$$\mathbf{R}\mathbf{G}_N^{(\mathbf{R})} = \mathbf{G}_N, \quad \mathbf{V}_\mathbf{R} = \mathbf{V}\mathbf{R}^T$$

That removes exactly those connectios with the two bits being both frozen or both infromation bits. And the linear constraints are modified accordingly.

Subcode Ensemble Decoding of Linear Block Codes



- Idea: See BLBC as union of a set of subcodes, decode parellelly and then ML combined.
- Construction of the set of subcodes $\mathbf{H}_\ell = \begin{bmatrix} \mathbf{H} \\ \mathbf{h}_\ell \end{bmatrix}$, $\mathbf{w}_H(\mathbf{h}_\ell) = d_c$
- Optimizing the subcode (of an LDPC code)
  - randomly generate arbitrary $\mathbf{h}_\ell$ while maintaining the code to have no four-cycles
  - Solved as the maximum coverage problem
- enable another parallelism decoding scheme for LDPC.

Subcode Ensemble Decoding of Polar Codes

Extending the subcode idea to polar code (that lacks such flexible tanner graph structure)

- Construction of the set of subcodes
  - Treat some of the info bits as dynamic frozen bit
- Subcode optimization
  - heuristically select the info bits with the least reliable bit-channels